Nitish Kumar

Senior Python Developer

Danapur,Patna,Bihar 801503 | +91 7543940627 | whereisnitish@gmail.com | https://www.linkedin.com/in/whereisnitish/

Education

Master of Computer Applications (MCA) - Artificial Intelligence & Machine Learning Chandigarh University

2021 - 2023 | CGPA: 7.7

Professional Summary

Innovative and analytical **Python Developer with 3+ years of experience**, specializing in **API-driven systems, AI voice agents, and backend development**. Strong expertise in building **scalable RESTful APIs**, AI-powered applications, and **LLM-integrated systems** using **Django, Flask, FastAPI**, and cloud platforms. Hands-on experience with **AI voice agents**, real-time communication workflows, and automation using **no-code/low-code platforms** such as **Make.com, n8n, and Zapier**.

Proficient in integrating **LLMs, NLP pipelines, and speech-based AI systems** into real-world products. Known for translating complex technical solutions into business value through strong client communication and cross-functional collaboration. Passionate about creating reliable, production-grade systems that bridge **AI innovation and practical use cases**.

Experience

Snakescript Solutions LLP

Senior Python Developer

April 2023 - Present

- Designed, developed, and deployed **API-first backend systems** using **Django, Flask, and FastAPI**, supporting AI-powered products and high-traffic client applications.
- Built and optimized **AI-driven platforms leveraging LLMs**, improving client response accuracy and reducing query response times by **25%**.
- Worked extensively on **AI voice agent workflows**, backend logic, and API orchestration for real-time interactions and call handling.
- Led client discussions, gathered requirements, and delivered AI-backed solutions, maintaining a **95% client satisfaction rate**.
- Led a Python development team using Agile practices, reducing delivery turnaround time by **30%**.
- Designed and optimized custom **machine learning and data processing pipelines**, improving efficiency by **60%** and model accuracy by **15%**.
- Reduced LLM API costs by **35%** through prompt optimization, caching strategies, and response validation.
- Developed scalable **microservices and RESTful APIs**, improving system modularity and reducing API latency by **40%**.
- Automated deployments and AI workflows using **GitHub Actions**, reducing deployment cycles by **50%**.
- Integrated **no-code automation tools (Make.com, n8n, Zapier)** for event-driven workflows, notifications, and system integrations.

Technical Skills

Programming Languages:

- Python, JavaScript

Machine Learning & AI:

- Frameworks: TensorFlow, PyTorch, Scikit-learn
- NLP: Transformers
- Computer Vision: OpenCV, MediaPipe
- LLMs: OpenAI API, RAG, Hugging Face Transformers
- AI Voice Agents & Conversational AI
- Agentic AI: Crew AI, LangGraph

No-Code / Low-Code Platforms

- Make.com, n8n, Zapier,Bubble.io

Web Development:

- **Backend**: Django, Flask, FastAPI,Nodejs
- **Frontend**: HTML, CSS, JavaScript

Databases:

- Relational: MySQL, PostgreSQL
- NoSQL: MongoDB

DevOps & Tools:

- Version Control: GitHub, Github Actions, JIRA
- Cloud: AWS (EC2, S3)

Data Analysis & Visualization:

- Pandas, NumPy, Matplotlib, Seaborn
- Jupyter Notebooks, PowerBI

Professional Strengths

- Strong analytical and problem-solving skills with a data-driven approach.
- Excellent communication abilities, adept at explaining complex concepts to non-technical stakeholders.
- Proactive learner, constantly updating skills in the rapidly evolving field of AI and ML.
- Thrives in fast-paced, collaborative environments with a focus on delivering high-quality results.

Notable Projects

Autonomous Browser Automation Agent (Playwright + MCP)

Tech Stack: Python, Playwright, MCP, noVNC, Llama (Open-Source 120B), OpenAI API, Redis

- Developed an AI-powered browser automation agent capable of navigating and interacting with web applications autonomously using natural language instructions.
- Implemented Model Context Protocol (MCP) integration to dynamically connect tools and manage structured communication between the LLM and browser environment.
- Built remote browser control using noVNC, enabling live monitoring and manual takeover of automated workflows.
- Integrated Llama 120B OSS model and OpenAI models for hybrid reasoning, improving task success rate and reducing automation failures.
- Designed site-specific automation logic for form filling, data extraction, and multi-step workflows across dynamic web pages.
- Implemented session memory and retry mechanisms, increasing automation reliability by 45%.
- Optimized execution using asynchronous Playwright operations, reducing workflow completion time by 30%.

AI-Powered Conversational Agent

Tech Stack: Python, OpenAI API, Django, Redis

- Developed an intelligent chatbot capable of understanding context and maintaining conversation history.
- Implemented custom logic to handle domain-specific queries, resulting in a 35% increase in successful query resolutions.

- Integrated text-to-speech and speech-to-text capabilities for enhanced accessibility.

TLC - Mobile App Backend Development

Tech Stack**:** Python, Django Rest Framework (DRF)

- Developed and optimized RESTful APIs for a cross-platform (Android/iOS) mobile application, ensuring seamless data exchange and high performance.
- Designed a scalable backend architecture, improving response times by 40% and handling concurrent requests efficiently.
- Implemented robust authentication and authorization mechanisms, enhancing app security and user data protection.
- Integrated database management and query optimization techniques, reducing API latency and improving system efficiency.
- Collaborated with frontend developers and QA teams to ensure smooth API consumption and flawless user experience.
- Deployed and maintained the backend system on a cloud infrastructure, enabling high availability and scalability.

E-commerce Platform with AI-Powered Recommendations

Tech Stack: Django, PostgreSQL, Stripe API, TensorFlow Recommenders

- Developed a fully functional e-commerce platform with user authentication, product management, and secure payment integration.
- Implemented a collaborative filtering recommendation system, increasing average order value by 20%.
- Optimized database queries and implemented caching, reducing page load times by 40%

Personal Projects

Personal Voice Assistant

[Talk to my Virtual Assistant]

Tech Stack: Python, Django, Vapi, Make.com, PythonAnywhere

Project Highlights:

- Built a **custom voice assistant** using **Vapi** that interacts with callers in real-time, handling conversations naturally and forwarding calls to me when needed.
- Automated **call summary emails** using **Make.com**, ensuring I receive detailed summaries after each interaction.
- Deployed the backend on **PythonAnywhere**, developed with **Python** and **Django**.

Hobbies

- **Chess & strategic games**
- **Badminton & fitness**
- **Reading & continuous learning**
- **Anchoring & public speaking**